UNIVERSAL SUBMERSIVENESS OF A REPRESENTABLE MORPHISM OF ALGEBRAIC SPACES TESTED BY BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays the square of topological spaces and uses only its openness, surjectivity and the surjectivity of the comparison map; the converse direction is stated as immediate from the definitions.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The description of the points of a fibre product of algebraic spaces. Properties of Spaces, Section spaces-properties-section-points, used at proof lines 1133-1145. Gives the commutativity and the open surjectivity properties of the square of topological spaces.
- Universal submersiveness in the representable sense. Section section-representable, used at proof lines 1119-1123 and 1130-1132. Fixes condition (1) and gives submersiveness of V x_Y X -> V.
- Surjectivity of the comparison map between the two fibre products of topological spaces. used at proof lines 1141-1145. Lets submersiveness pass from the left vertical arrow to the right one.

Source reference. Original section 03HA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4559–4559. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 03BT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4360–4360. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a representable morphism of algebraic spaces over S. The following are equivalent*

(1) *f is universally submersive (in the sense of Section 03HA), and*
(2) *for every morphism of algebraic spaces $Z \to Y$ the morphism of topological spaces $|Z \times_Y X| \to |Z|$ is submersive.*

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Assume (1), and let $Z \to Y$ be as in (2). Choose a scheme V and a surjective étale morphism $V \to Y$. By assumption the morphism of schemes $V \times_Y X \to V$ is universally submersive. By Properties of Spaces, Section 03BT in the commutative diagram

$$\begin{array}{ccc} |V \times_Y X| & \longrightarrow & |Z \times_Y X| \\ \downarrow & & \downarrow \\ |V| & \longrightarrow & |Z| \end{array}$$

the horizontal arrows are open and surjective, and moreover

$$|V \times_Y X| \longrightarrow |V| \times_{|Z|} |Z \times_Y X|$$

is surjective. Hence as the left vertical arrow is submersive it follows that the right vertical arrow is submersive. This proves (2). The implication (2) $\Rightarrow$ (1) is immediate from the definitions. $\square$